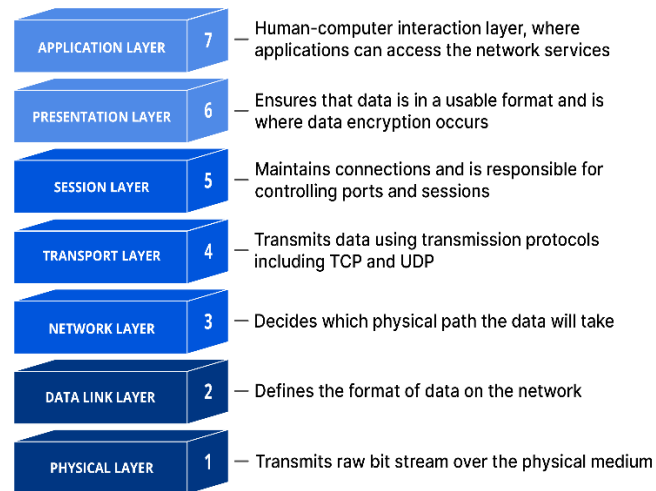


OSI Layer (Open Systems Interconnection) and Protocols



Layer 7 – Application

HTTP / HTTPS

Layer 6 – Presentation

SSL / TLS Encryption

Layer 5 – Session

NetBIOS / RPC

Layer 4 – Transport

TCP / UDP

Layer 3 – Network

Router / IP

Layer 2 – Data Link

Switch / Ethernet

Layer 1 – Physical

Cables / Hub / Signals

2. Match the Scenario to the Correct OSI Layer

- a) Encrypting data before sending: Presentation Layer (Layer 6)
- b) Routing packets across networks: Network Layer (Layer 3)
- c) Converting data to electrical signals: Physical Layer (Layer 1)
- d) Opening a web page in your browser: Application Layer (Layer 7)
- e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

3. Example Of WhatsApp

Suppose I send a message on WhatsApp.

Layer	Working
Application (7)	WhatsApp creates the message and displays it in the app.
Presentation (6)	The message is encrypted using WhatsApp's end-to-end encryption.
Transport (4)	TCP ensures the message reaches the server reliably.
Network (3)	IP packets are routed through the internet to the recipient.
Data Link (2)	Wi-Fi or mobile network frames are sent between my phone and the router/tower.
Physical (1)	Radio waves or electrical signals carry the data physically.

4. Troubleshooting Scenario

Problem: You cannot open a website, but you can successfully ping its IP address.

What this tells us:

- Ping works → Network Layer (Layer 3) is functioning. The IP address is reachable, routing is working, and the host is responding.
- Website does not open → Problem is likely above Layer 3.

Most likely layers causing the issue:

- Application Layer (Layer 7): The web service (HTTP/HTTPS) may be down or blocked.
- Presentation Layer (Layer 6): SSL/TLS certificate or encryption problems may prevent HTTPS from loading.
- Session Layer (Layer 5): The browser may fail to establish or maintain a session with the web server.

Reason

Because ICMP ping succeeds, Layers 1–3 are working correctly. The failure occurs when the browser tries to use HTTP/HTTPS, which operates mainly at Layer 7 (with Layer 6 involved for HTTPS encryption). Therefore, the problem is most likely in the Application, Presentation, or Session layer, not in the Physical, Data Link, or Network layers.